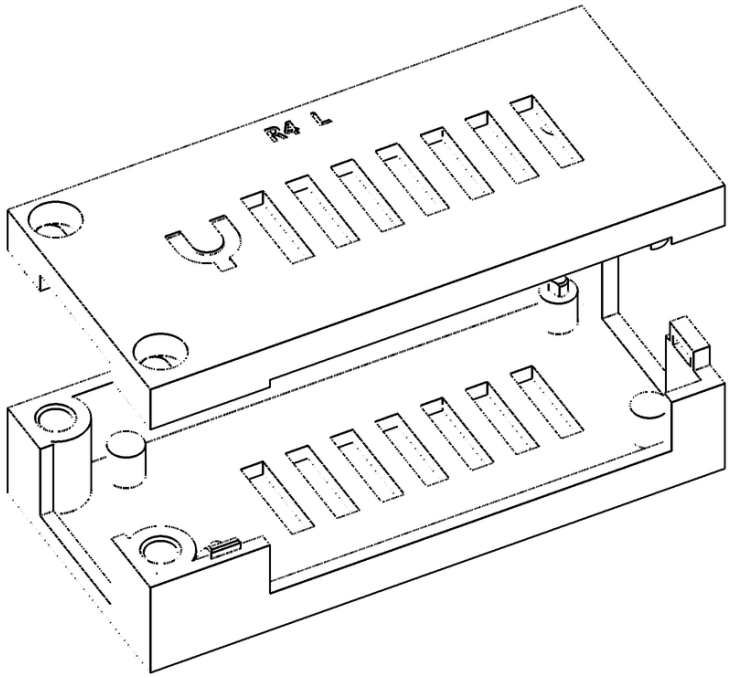
Inventory, orientation and board placement

R4 B base + R4 L lid. Flat rectangle: 76.8 x 35.9 x 20.0 mm.
Two M3 x 8 socket-head screws from the same FGRUH kit used by Lacuna.
Two RX M3 x 5.7 heat-set inserts: catalogue OD 4.6 / length 5.7 / pilot 4.0 mm.
Stock identity is unconfirmed. Use the maker profile only after checking the label.
Nominal screw envelope: head <= OD 6 x height 3 mm; check actual hardware.
Tools: caliper/depth gauge, M3 socket key, controlled insert heater, padded mat,
actual service Ethernet + both USB-C cables, 0-10 N gauge, timer/test host.

LID: three guide slots, two rear retaining tabs, two recessed heads



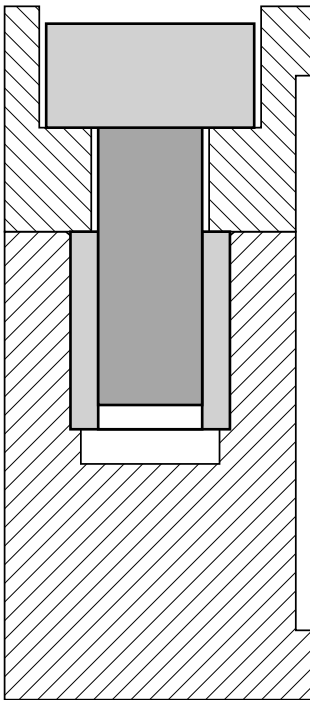
BASE: two insert bosses beside RJ45; round + relieved PCB pegs

1. Qualify the small joints using coupon-assembly.pdf before a full case.
2. Heat-set inserts in the EMPTY base before any board or lid is installed (page 2).
3. Unpowered bare ZERO 3E V1.2: RJ45 faces -X / left, USB-C faces -Y / front.
   Lower board vertically and parallel onto four pads. Align the round peg first,
   then the diagonal relieved peg. Never force a connector, pin, or component.
4. Test lid closure without screws as on page 3; it must seat and slide by hand.

No screw passes through a PCB hole. Original PCB bores are OD 2.75 mm.
Both parts print exterior face down. Open bores; zero sacrificial membranes.

Insert installation and exact screw section

A-A at Y = 1.6 mm: actual base/lid section; X horizontal, Z up.
Opposite hatching distinguishes printed parts; gray is nominal metal envelope.



z 20.0 exterior top

z 19.5 nominal head top: recess 0.50

z 16.5 head bearing land

z 13.5 boss hard stop / insert top

z 8.5 8 mm screw tip

z 7.8 provisional insert bottom

z 6.8 blind pilot bottom

Nominal grip 3.0; axial overlap 5.0 mm.

Tip-to-insert-bottom reserve 0.70 mm.

Tip-to-plastic-bottom reserve 1.70 mm.

OD 8.4 boss: 1.9 mm around OD 4.6 insert.

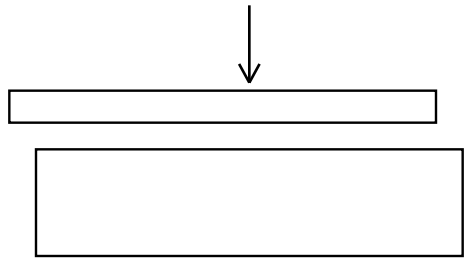
OD 6.4 head well: 1.0 mm radial wall.

Axes: (-5, 1.6) and (-5, 28) mm.

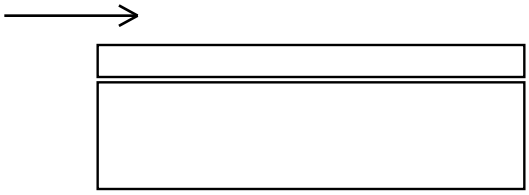
HEAT-SET ACTION (before board): support base flat, hold heater normal to boss.
Set each insert flush at Z 13.5, keep its bore coaxial, let it cool fully before loading.
Tool envelope: OD 2.4 pilot x 1, OD 4.2 shoulder x 2, then OD 5 body. Heater must fit with pilot 1 mm engaged; use the maker heat process.
Do not install into a guessed pilot, press cold, or use the PCB as a support.
SCREW ACTION (after free slot closure): insert two M3 x 8 downward from top.
OD 3 driver shaft reaches each head vertically through the OD 6.4 open well.
Start both threads by hand; lightly snug alternately. No qualified torque exists.
Load path: head -> lid boss -> base boss hard stop -> embedded metal insert.
Screw preload bypasses the PCB; upper pads leave nominal 0.18 mm capture play.
Measured acceptance: nominal axial overlap >= 4.5 mm; tip reserve >= 0.4 mm; head recess >= 0.2 mm. Do not substitute a longer screw or tighten away interference.

Slot together, then screw together

5. Hold lid level 1.2 mm LEFT (-X) of the base. Keep all cables disconnected.
   Lower vertically: align three chamfered tongues with their elongated slots.
6. Slide lid RIGHT (+X) 1.2 mm until all outside faces align. Do not force it.
   Both rear tabs now sit under the receiver roofs; only then fit the screws.

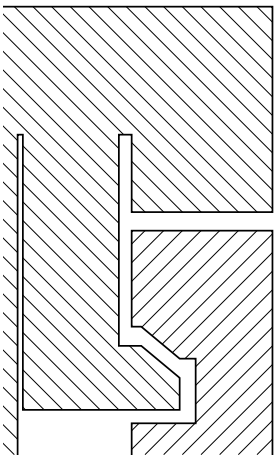


5 LOWER AT X OFFSET -1.2



6 SLIDE +1.2 TO FLUSH

B-B rear tab at Y = 3.0 mm (X / Z); exact STL contours, 24x scale.



Tab overlap under roof: 0.75 mm.

Tab root / tip: 1.0 / 0.5 mm.

Pocket roof thickness: 1.5 mm.

Remaining rear wall: 1.2 mm.

Matching roof chamfer: 0.30 vertical clearance.

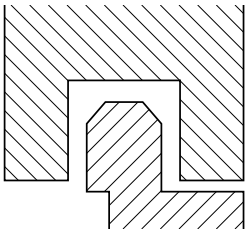
Pocket width 4.4; tab 3.9; side gap 0.25.

Rear slots prevent straight vertical lift.

Screws prevent the reverse slide.

Physical retention strength: NOT RUN.

C-C guide at X = 16 mm (Y / Z): tongue 1.0 wide x 1.2 high, 0.3 lead-in.



Slot width 1.5; tongue side clearance 0.25.
Guide slot has 0.85 mm material each side.
Extra X length permits the 1.2 mm slide.
No snap flexure or glue is required.
Front-right board pad is relieved to clear
the USB-C shell during that slide.

The three guides carry XY location. Rear tabs carry opening load at the rear.
Front screws clamp the hard stops; no external ear or turret exists.

Fit tests, service and first-article decisions

STATUS: every physical observation below is NOT RUN. Digital fit is provisional. Before release: confirm actual insert label/OD/length; measure both screws (length, head OD/height), actual board thickness and actual heater/driver envelope. Regenerate exact parts/plate/PDF, review every page and joint independently, then perform the normal Lacuna exact-job release. No R1, R2 or R3 approval transfers.

Unpowered fit and retention

A. Measure empty parts. Require clear bores, intact flat seats and no cracks/sag. Fit inserts only after pilot confirmation. Reject any boss whitening or looseness.

B. Close empty case WITHOUT screws: lower at -1.2 X, slide +1.2 X by hand. Require flush exterior <= 0.2 mm mismatch, no forced bend, all three guides seated and rear hooks preventing a straight lift. Reopen by reversing the slide.

C. Fit the board without cables or screws. It must sit flat on all four pads. Require no contact with components, no bowed PCB, and free repeat closure.

D. Check measured stack: >= 4.5 mm axial overlap, >= 0.4 mm tip reserve, >= 0.2 mm head recess. These limits include print error; never tighten to fix fit.

E. Start both M3 x 8 screws and lightly snug. Rotate over a padded mat through six orientations for 10 s: no release; board play <=0.3 mm.

F. Use actual Ethernet and both USB-C cables. Require full engagement, operable latches and straight removal without touching plastic. Hold case, not PCB/ports.

G. With board unpowered and cables removed, apply 5 N separating load between halves for 10 s each at front/rear, over padded mat. Require no release, crack or permanent gap > 0.2 mm. Repeat assembly/service 10 cycles and reinspect.

Powered observation and reverse service

H. After unpowered checks pass, run intended workload for 30 min at measured ambient <= 30 C. Record CPU temperature once/minute, throttling and resets. Require no reset, thermal throttling, plastic distortion or CPU reading >= 75 C. This is a trial limit, not a rated thermal envelope.

SERVICE: unplug all power/cables; support base on mat. Remove both screws. Slide lid LEFT 1.2 mm before lifting. Lift level; lift PCB vertically off pegs by edges. Inserts stay in base. If an insert turns, cracks or withdraws, stop and revise boss.

STOP for forced fit, bowed PCB, damaged connector, stalled screw, bottoming, insert motion, crack, heat, smoke, smell, or any failed limit. Record dimensions and the failed interface; return to design. Physical acceptance needs observations.

Limits: bare board only; no PoE HAT/heatsink; no ingress or torque rating. Board surface-edge and sampled checks can miss collisions between samples.